FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2022

TEST DATE: 17-02-2022

ANSWERS, HINTS & SOLUTIONS

Physics PART – A

SECTION – A

1. B

2. A

3. C

Sol. From conservation of angular momentum about hinged point

$$mv\ell = \left(\frac{m\ell^2}{3} + 2m.\ell^2\right)\omega$$

$$\omega = \left(\frac{3}{7}.\frac{v}{\ell}\right)$$

Now from conservation of energy

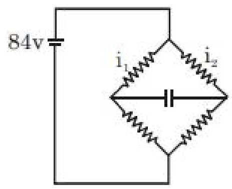
$$\frac{1}{2}\times\frac{7}{3}m\ell^2\times\omega^2 = mgl + 2mg.2\ell$$

$$v = \sqrt{\frac{70}{3}g\ell}$$

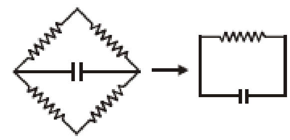
4. C

Sol. $\mathrm{i}_1 = \frac{84}{140} = 0.6A$

$$i_2 = \frac{84}{60} = 1.4A$$



p.d across capacitor = 20 v
Q = CV = 5 × 20 = 100 μC
After S is open $Q = Q.e^{-t/RC}$
$= 100e^{-1}$



5. D
Sol. Stopping potential increases and number of photons decreases.
$$I = \frac{hcn_p}{\lambda t}$$

6. A

7. B
Sol. $$\frac{kq'}{R} + \frac{kq}{2R} = 0$$
$$q' = -\frac{q}{2}$$
So charge flown in to the earth $+\frac{q}{2}$

8. C
Sol. The time period of LC oscillations, $T = 2\pi\sqrt{LC}$
The time at which charge on the capacitor will be zero is $T/4$
So $t = \frac{\pi}{2}\sqrt{LC}$

9. A
Sol. $$I = \frac{E}{R_2}$$
$$U_{inductor} = \text{Heat loss in } R_1 = \frac{1}{2}L\left(\frac{E}{R_2}\right)^2$$

10. D
Sol. Using energy method:
$$E = \frac{1}{2}mv^2 + \frac{1}{2}\frac{MR^2}{2}\frac{v^2}{R^2} + \frac{1}{2}kx^2$$

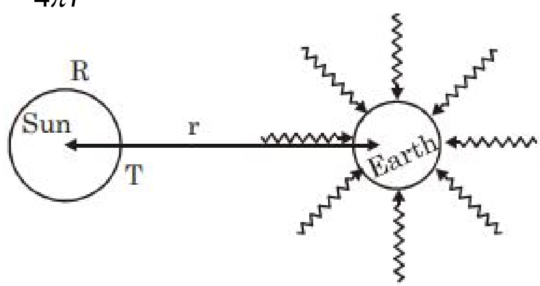
$$E=\frac{3}{4}mv^2+\frac{1}{2}kx^2$$

$$\frac{dE}{dt}=0=\frac{3}{2}Mv.a+kxv$$

As $V\neq 0 \Rightarrow a=-\frac{2}{3}\frac{k}{M}x \Rightarrow T=2\pi\sqrt{\frac{3M}{2k}}$

11. A

Sol. $\frac{eA_{sun}\sigma T^1}{4\pi r^2}\times A_{proj\,area}$



$$\frac{4\pi R^2}{4\pi r^2}\sigma T^4\pi r_0^2$$

$$\frac{\pi r_0^2 R^2\sigma T^4}{r^2}$$

12. A

13. A

14. D

Sol. $C_p=\frac{Q}{n\Delta\theta}=\frac{70}{2\times 5}=7cal\,mol^{-1}K^{-1}$

$C_v=C_p-R=7-2=5cal\,mol^{-1}K^{-1}$

$Q=nC_v\Delta\theta=2\times5\times5=50cal$

15. A

Sol. $\beta=\frac{\lambda D}{d}=0.25\times10^{-3}\Rightarrow \lambda D=\frac{d}{4}\times10^{-3}....(1)$

$\beta'=\frac{\lambda D}{d+1.2\times10^{-3}}=\frac{\beta}{1.5}=\frac{0.25\times10^{-3}}{1.5}=\frac{10^{-3}}{6}\Rightarrow \lambda D=\frac{10^{-3}}{6}\left(d+1.2\times10^{-3}\right)....(2)$

From equation = $(1)\text{and}(2), \frac{d}{4}=\frac{d}{6}+\frac{1.2\times10^{-3}}{6}\Rightarrow d\left(\frac{1}{4}-\frac{1}{6}\right)=\frac{1.2\times10^{-3}}{6}$

$\Rightarrow d=2.4\times10^{-3}m$

From (1): $\lambda=\frac{d\times10^{-3}}{4D}=\frac{2.4\times10^{-3}\times10^{-3}}{4\times1}=0.6\,\mu m$

16. C

Sol. $T = 2\pi\sqrt{1/g}$

If effective acceleration due to gravity $= g - 3g/4 = g/4$

New time period $= T' = 2\pi\sqrt{1/g/4}$

$= 2\left[2\pi\sqrt{1/g}\right] = 2T$

17. B

18. B

Sol. $\mathrm{E}\alpha\mathrm{AT}^{1}$

$\mathrm{E}\alpha r^{2}$

$\therefore \frac{E_1}{E_2} = \frac{r_1^2}{r_2^2} = \left(\frac{2}{1}\right)^2 = \frac{4}{1}$

19. D

Sol. $\beta_1 = 10\log_{10}\left(\frac{\mathrm{I}}{\mathrm{I}_0}\right) and\ \beta_2 \log_{10}\left(\frac{I'}{I_0}\right)$

Given, $\beta_1 - \beta_2 = 20$

$\therefore 20 = 10\log\left(I/I'\right)$

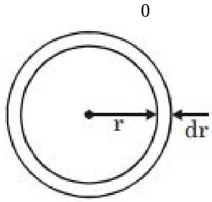
$\Rightarrow I'\frac{I}{100}$

20. A

SECTION – B

21. 00000.63

Sol. $\tau = \int F.r = \int_0^R \eta\left(2\pi r\, dr\right)\left(\frac{rw}{t}\right).r$



$= \frac{2\pi\eta WR^4}{4t} = \frac{2\times 3.14\times 1\times 8\times 10^{-4}}{4\times 2\times 10^{-3}}$

$= 0.625N - m = 0.63N - m$

22. 00011.70

Sol. $\mathrm{I} = \frac{12-0.3}{5\times 10^3} = 2.34mA$

$\mathrm{V}_0 = \mathrm{IR} = \left(2.34\times 11^{-3}\right)\left(5\times 10^{3}\right) = 11.7\mathrm{V}$

23. 00002.00

Sol. $n(7800)=(n+1)5200$

24. 00000.40

Sol. $\Delta L = L_0\,\alpha\,\Delta T$

$=12\times 11\times 10^{-6}\times 30$

$=3960\times 10^{-6}\,m$

25. 00000.09

Sol. Energy in 1 loop $\int \frac{1}{2}dm\left[A\sin(kx)\omega\right]^2$

$=\frac{\lambda A^2\omega^2}{8}(M/L)$

$\therefore$ Ratio $=\frac{3}{5}\times\frac{3}{5}\times\frac{1}{4}$

26. 00001.00

Sol. Centre of mass of binary star lies at centre of their circular path.

$M_1r_1 = M_2r_2$

Then $G=\frac{M_1M_2}{(r_1+r_2)^2}=M_1W_1^2r_1=M_2W_2^2r_2$

$\Rightarrow w_1 = w_2 \rightarrow \frac{T_1}{T_2}=1.00$

27. 00003.75

Sol. From equation of motion $v^2=u^2+2as$

$\Rightarrow (5000)^2=(1000)^2-2as \Rightarrow s=\frac{37500}{a}$

$=ma\times s=\frac{10}{1000}\times 375000$

$=3750\,Joule = 3.75\,kJ$

28. 00001.05

29. 00000.45

Sol. Induced emf

$=eB_Hlv=5\times 10^{-5}\times 3\times 3=0.45\times 10^{-3}=0.45\,mV$

30. 00001.10

Chemistry

PART – B

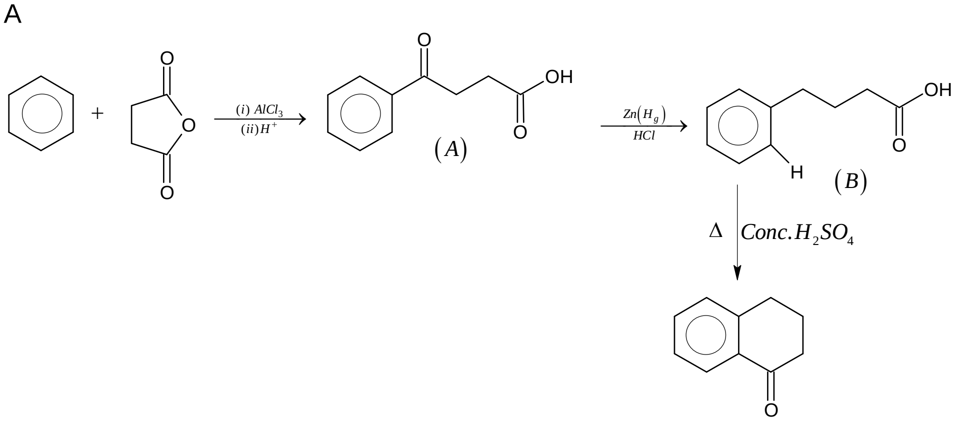
SECTION – A

31. C
Sol. Ionic mobilities of hydrated alkali metal ions are as:
$Cs^{+} > Rb^{+} > K^{+} > Na^{+} > Li^{+}$

32. C
Sol. Macromolecular colloids like starch, gelatin are generally lyophilic.

33. A
Sol. Groups to be eliminated should be trans-biaxial in chair conformation.

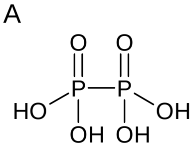
34. A
Sol.



35. B
Sol. $\underset{(A)}{Bi^{3+}} \xrightarrow{KI} \underset{(B)}{BiI_3} \xrightarrow[KI]{excess} [BiI_4]^{-}$

Black ppt yellow brown solution

36. A
Sol.



$H_4P_2O_6$

37. C
Sol. $AgBr = x+y$

$$AgBr = \frac{K \times 1000}{M}$$

M – molarity of $AgBr$ solution

8 (solubility in g/L) = $\dfrac{K \times 1000}{(x+y)} \times 188$

38. D

Sol. Equivalents of $\mathrm{KMnO_4} = 5x$

Equivalents of $\mathrm{K_2Cr_2O_7} = 6y$

Equivalents of oxidising agent must be same $5x = 6y$

$x : y = 6 : 5$

39. B

Sol. Greater the activation energy of a reaction, greater is the temperature dependence of rate constant of reaction.

40. C

Sol. $\frac{1}{\lambda} = RZ^2\left(1 - \frac{1}{4}\right)$ for lyman line

41. C

Sol. Cl_2 is present simultaneously in two equilibria. Decrease in its concentration shift both equilibria forward.

42. D

Sol. Fluorine is most reactive amongst halides and has highest oxidising power

43. A

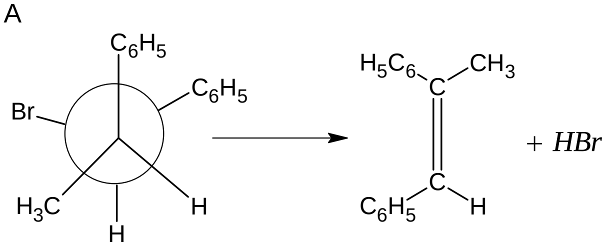
Sol. $\underset{sp^3d}{\left[XeF_3\right]^+}\ \underset{sp^3d^2}{\left[SbF_6\right]^-}$

44. D

Sol. In $\mathrm{PCl_2F_3}$, net dipole moment does not cancel out.

45. A

Sol.

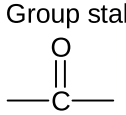


46. C

Sol. $-\mathrm{OH}$ group of phenol cannot be replaced by iodine as it does not undergo nucleophilic substitution reaction under normal conditions.

47. A

Sol. Group stabilise the carbanion character on $\alpha-$carbon



48. C
Sol. Methoxy group stabilises the cationic intermediate by $+\mathrm{M}$ effect

49. B
Sol. 2 molecules of RMgX are consumed by acid chloride group as it undergoes addition elimination first and then addition. The other aldehydic & ketonic group consume one RMgX molecule each.

50. B
Sol. Structure containing a ring of $(4n+2)\pi e^{-}s$ is aromatic

SECTION – B

51. 00002.00
Sol. $\mathrm{Ca_3P_2+6H_2O \longrightarrow 3Ca(OH)_2+2PH_3}$

52. 00001.90
Sol. $\Delta\mathrm{U=3\times1.5R\times100+2\times2.5R\times100=1900\,cal}$
$=1.9\,\mathrm{K\,cal}$

53. 00011.00
Sol. 542 and sum of digits = 5+4+2 = 11

54. 00006.00
Sol. $\mathrm{CrO_2Cl_2}$

55. 00454.00
Sol. $\mathrm{C_2H_5I(156), C_2H_5I(156), CH_3I(142)}$

56. 00009.00
Sol. $\Delta\mathrm{H}=\mathrm{BE_R}-\mathrm{BE_p}$
$=0-[36\mathrm{BE}_{B-B}]$
$=-36\times200=-7200$
$|\Delta\mathrm{H}|=7200$
Sum of digits = 9

57. 00003.00
Sol. $\mathrm{Cu^{2+},\ Sc^{2+}, NO_2}$

58. 00012.00
Sol. $\kappa=\kappa_{[M(NH_3)_4Br_2]^{+1}}+\kappa_{H_2PO_2^{-1}}$
$\frac{1}{200}=\frac{400\times10^4 s}{1000}+\frac{100\times10^4 s}{1000}$
$s=10^{-6}$
$K_{sp}=10^{-12}, x=12$

59. 00003.00

Sol. Ibuprofen, Aspirin and Phenacitin are antipyretics

60. 00008.00

Sol. (i) give NO_2

(ii) NH_4NO_3

(iii) gives NO_2

(iv) give NO_2

(v) give NO_2

(vi) given N_2O

(vii) gives NO_2

(viii) gives NO_2

(ix) gives NO

Mathematics

PART – C

SECTION – A

61. A

Sol. $\sin 47° - \sin 25° + \tan 61° - \sin 11°$

$= (\sin 61° + \sin 47°) - (\sin 25° + \sin 11°)$

$= 2\sin 54° \cos 7° - 2\sin 18° \cos 7°$

$= 2\cos 7° (\sin 54° - \sin 18°)$

$= 2\cos 7° (\cos 36° - \sin 18°)$

$= 2\cos 7° \left(\frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4} \right) = \cos 7°$

62. C

Sol. Let equation of tangent for $\left(ct, \frac{c}{t} \right)$ is $ty + \frac{x}{t} - 2c = 0$

Normal from P is $xt^3 - ty - ct^4 + c = 0$

Co – ordinate of T $(2ct, 0)$, $T'\left(0, \frac{2c}{t} \right)$, $N\left(\frac{ct^4 - c}{t^3}, 0 \right)$ and $N'\left(0, \frac{c - ct^4}{t} \right)$

$\Rightarrow \Delta = \frac{c^2(1+t^4)}{2t^4}$ and $\Delta' = \frac{c^2}{2}(1+t^4)$

$\frac{1}{\Delta} + \frac{1}{\Delta'} = \frac{2}{c^2}$

63. B

64. A

65. C

66. A

67. B

68. A

69. D

Sol. The point $([P+1],[P])$ lies inside the circle $x^2 + y^2 - 2x - 15 = 0$

$\therefore [P+1]^2 + [P]^2 - 2[P+1] - 15 < 0$

$\Rightarrow \quad [P]^2 < 8$ .... (1)

Point $([P+1].[P])$ lies outside the circle $x^2 + y^2 - 2x - 7 = 0$

$\Rightarrow [P+1]^2 + [P]^2 - 2[P+1] - 7 > 0$

$\Rightarrow [P]^2 > 4$ .... (2)

$4 < [P]^2 < 8$

70. B

Sol. Centres of circles are

$C_1(-\lambda,0), C_2(-\lambda_2,0), C_3(-\lambda_3,0)$

$\because OC_1, OC_2, OC_3$ are in GP

$\therefore \lambda_2^2 = \lambda_1, \lambda_3$

Let any point on $x^2 + y^2 = c^2$ be $P(c\cos\alpha, c\sin\alpha)$,

$\therefore$ Length of tangents are

$PT_1 = \sqrt{2\lambda_1(c\cos\alpha)}$

$PT_2 = \sqrt{2\lambda_2(c\cos\alpha)}$

$PT_3 = \sqrt{2\lambda_3(c\cos\alpha)}$

$\Rightarrow PT_2^2 = PT_1, PT_3$

71. C

Sol. Equation of the radical axis is

$4\lambda x = 9 \quad or \quad x = \frac{9}{4\lambda}$

Pitting x in $x^2 + y^2 = 1$

$y = \pm\sqrt{1 - \frac{81}{16\lambda^2}}$

For the circle to have two common tangents, they must intersect at two distinct points

$\therefore \quad 1 - \frac{81}{16\lambda^2} > 0$

$\Rightarrow \lambda < \frac{-9}{4} \quad or \quad \lambda > \frac{9}{4}$

72. C

Sol. $y = |x-1|$

$\Rightarrow y^2 = (x-1)^2$

For image in y- axis, charge x by $-x$

$y^2 = (-x-1)^2$

$\Rightarrow x^2 - y^2 + 2x + 1 = 0$

73. D

Sol. Use $\Delta = 0$

$16\gamma^2 - 16p^2\gamma + 4p(\alpha p + \beta p - \alpha\beta) = 0$

Since γ is real

$\therefore D \geq 0$

$\Rightarrow (p-\alpha)(p-\beta) \geq 0$

74. A

Sol. $\because \frac{1}{10!}\left\{\frac{2\times 10!}{1!9!}+\frac{2\times 10!}{3!7!}+\frac{10!}{5!5!}\right\}=\frac{2^m}{n!}$

$\Rightarrow \frac{1}{10!}\left\{{}^{10}C_1+{}^{10}C_3+.......+{}^{10}C_9\right\}=\frac{2^m}{n!}$

$\Rightarrow \frac{1}{10!}2^9=\frac{2^m}{n!}$

$\therefore m=9\ n=10$

$\therefore$ Lines $x-y+1=0\ \&\ x+y+3=0$ are perpendicular to each other

$\therefore$ Orthocentre $(-2,-1)$

$\therefore (2m-2n, m-n)$

75. B

Sol. Since $\tan\left(\frac{B-C}{2}\right)=\frac{b-c}{b+c}\cos\frac{A}{2}$ $[\tan gent\, rule]$

$\therefore \tan\left(\frac{B-C}{2}\right)\cot\left(\frac{B+C}{2}\right)=\frac{b-c}{b+c}$

$\Rightarrow p=\frac{b-c}{b+c}$ $\therefore \frac{1+p}{1-p}=\frac{b}{c}-----(1)$

Similarly $\frac{1+q}{1-q}=\frac{c}{a}----(2)$

$\frac{1+p}{1-r}=\frac{a}{b}---(3)$

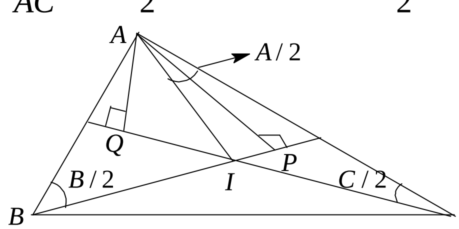
$\therefore \left(\frac{1+p}{1-p}\right)\left(\frac{1+q}{1-q}\right)\left(\frac{1+r}{1-r}\right)=1$

$\Rightarrow 2(p+q+r)=-2pqr$

$\sum p=-pqr$

76. D

Sol. $\frac{AQ}{AC}=\sin\frac{C}{2}$ $\Rightarrow AQ=AC\sin\frac{C}{2}.......(1)$



In ΔAIC, $\dfrac{CI}{\sin\dfrac{A}{2}} = \dfrac{AC}{\sin\left\{\pi - \left(\dfrac{A+C}{2}\right)\right\}}$

$\Rightarrow CI = AC\dfrac{\sin A/2}{\cos B/2} ----(2)$

$(1)/(2)\ \ \dfrac{AQ}{CI} = \dfrac{\cos\dfrac{B}{2}\sin\dfrac{C}{2}}{\sin\dfrac{A}{2}}$

Similarly $\dfrac{AP}{BI} = \dfrac{\cos\dfrac{C}{2}\sin\dfrac{B}{2}}{\sin\dfrac{A}{2}}$

$\dfrac{AQ}{CI} + \dfrac{AP}{BI} = \dfrac{\sin\left(\dfrac{B}{2}+\dfrac{C}{2}\right)}{\sin\dfrac{A}{2}}\ \ = \tan\left(\dfrac{B+C}{2}\right)$

77. B

78. A

Sol. Putting n = 1 $P(1) = 7^2 + 2^\circ.3^\circ = 50 = 2\times 25$

Also confirm by $P(n) \Rightarrow P(n+1)$ induction step

79. D

Sol. $n(n+1)$ is always even. So given $P(n)$ is wrong.

80. A

SECTION – B

81. 00002.00

Sol. $4x^2 - 4x + 2 = \sin^2 y$

$\text{since}\, 0 \le \sin^2 y \le 1$

$\Rightarrow 0 < 4x^2 - 4x + 2 \le 1 \quad (\because 4x^2 - 4x + 2 > 0)$

$\Rightarrow (2x-1)^2 \le 0$

$\Rightarrow x = \dfrac{1}{2}$

$\therefore \sin^2 y = 1$

$y = \pm\dfrac{\pi}{2}$

82. 00004.00

83. 00004.00

84. 00002.00

85. 02011.25

Sol. dividend= x^{2007},

Divisor= x^2-5x+6

Let Q(x)be quotient, R(x)=ax+b be remainder

$\mathrm{x}^{2007}=\mathrm{Q(x).(x^2\text{-}5x+6)+ax+b}$

$\mathrm{x}^{2007}=\mathrm{Q(x).(x\text{-}2)(x\text{-}3)}+\underbrace{\mathrm{ax+b}}_{R(x)}$ .....(i)

Now R(0)=b=?

Put x=2 in Eq. (i)

$2a+b=2^{2007}$ ....(ii)

Put x=3 in Eq. (i)

$\mathrm{3a+b=3^{2007}}$ ....(iii)

Eq. (iii)-Eq.(ii) gives

$a=3^{2007}-2^{2007}$

Now $\mathrm{b=2^{2007}\text{-}2a=2^{2007}\text{-}2(3^{2007}\text{-}2^{2007})=2^{2007}+2.2^{2007}\text{-}2.3^{2007}}$

$\mathrm{=2.3[2^{2006}\text{-}3^{2006}]=ab(a^c\text{-}b^c)}$

Hence a=2;b=3;c=2006

86. 00004.00

Sol. $$\cos^2 x+\left(\frac{\sqrt{3}+1}{2}\right)\sin x-\frac{\sqrt{3}}{4}-1=0$$

$$\Rightarrow 1-\sin^2 x+\left(\frac{\sqrt{3}+1}{2}\right)\sin x-\frac{\sqrt{3}}{4}-1=0$$

$$\Rightarrow \left(2\sin x-\sqrt{3}\right)\left(2\sin x-1\right)=0$$

$$\Rightarrow x=\frac{\pi}{3},\frac{2\pi}{3},\frac{\pi}{6},\frac{5\pi}{6} \quad \because x\in[-\pi,\pi]$$

87. 00028.00

Sol. $$\tan\left(5\pi\cos\theta\right)=\tan\left\{\frac{\pi}{2}-5\pi\sin\theta\right\}$$

$$\Rightarrow 5\pi\cos\theta=n\pi+\left(\frac{\pi}{2}-5\pi\sin\theta\right)$$

$$\Rightarrow \cos\theta+\sin\theta=\frac{2n+1}{10}$$

$$\Rightarrow \sin\left(\theta+\frac{\pi}{4}\right)=\frac{2n+1}{10\sqrt{2}}$$

$n=14,$ for each n thane are two values of θ

$\therefore 28\, solutions$

88. 00002.00

Sol. Let $\tan x = t$

$\therefore$ Equation reduces to

$$\frac{4t^2}{1+t^2}+t^2+\frac{2}{t^2}=5$$

$$\Rightarrow \left(t^2-1\right)\left(t^4+t^2-2\right)=0$$

$$\Rightarrow t^2=1,-2$$

$$x=\frac{\pi}{4},\frac{3\pi}{4}$$

89. 00005.00

Sol. First note that out of the 8 selected cards, one pair of cards have to share the same number and another pair of cards have to share the same colour. Now, these 2 pairs of cards cannot be the same or else there will be 2 cards which are completely same.

Then WLOG let the numbers be $1,1,2,3,4,5,6$ and 7 and the colours be $a,a,b,c,d,e,\ f$ and g. We therefore obtain only 2 cases:

Case I: $1a\,1b\,2a,3c,4d,5e,6f,7g$. In this case, we can discard $1a$

There are $2\times 6=12$ situation in this case

Case II: $1b,1c,2a,3a,4d,5e,6f$ and $7g$. In this case, we cannot discard. There are ${}^{6}C_2=15$ situations in this case.

Probability $=\dfrac{12}{12+15}=\dfrac{4}{9}\therefore \left|p-q\right|=5$

90. 00001.00

Sol. $$f\left(y\right)=e^y\lim_{n\to\infty}\sum_{n=1}^{n}\left(\frac{\left(e^{n+1}-1\right)-\left(e^n-1\right)}{\left(e^n-1\right)\left(e^{n+1}-1\right)}\right)$$

$$=e^y\lim_{n\to\infty}\sum_{n=1}^{n}\left(\frac{1}{e^n}-\frac{1}{e^{n+1}-1}\right)$$

$$=\frac{e^y}{e-1}$$

$$\therefore \int_0^1 f\left(x\right)dx=\frac{1}{e-1}\int_0^1 e^x dx=1$$